



Bridge Inspection Report

09698

Team Leader: Colin Redington

Inspection Date: 11/13/2024

(7) Facility Carried: SC 22 EB

(6) Crossing: JONES BIG SWAMP

Inspection Type(s): Routine

(16) Latitude: 33.893483 (17) Longitude: -78.852736

IDENTIFICATION

Asset ID:	09698	(009) Location [B.L.11]:	I N BRIARCLIFF ACRES
(468) Structure Number:	2640002221300	(041) Traffic Status: County [B.PS.01]:	A - Open, no restriction
(002) District [B.L.04]:	District 5	(003) County [B.L.02]:	(26) Horry
(42A) Type of Service On [B.F.01]:	1 Highway	(004) Placecode [B.L.03]:	Unknown
(006) Feature Intersected:	JONES BIG SWAMP	(022) Owner [B.CL.01]:	SCDOT
(42B) Type of Service Under [B.F.02]:	5 Waterway	(011) Mile Post [B.H.07]:	19.887 mi (32.005 km)
(021) Custodian:	01 SCDOT	(106) Year Recon [B.W.02]:	0
(027) Year Built [B.W.01]:	2000	(037) Historical:	5 Not eligible for NRHP

CONDITION AND APPRAISAL

(058) Deck Structure Condition [B.C.01]:	7 Good	(602) Bridge Railing Condition [B.C.05]	7 Good
(059) Superstructure Condition [B.C.02]:	7 Good	(603) Bridge Railing Trans. Condition [B.C.06]:	6 Satisfactory
(060) Substructure Condition [B.C.03]:	6 Satisfactory	(604) Bridge Bearing Condition [B.C.07]	8 Very Good
(062) Culvert Retaining Condition [B.C.04]:	N N/A (NBI)	(605) Bridge Joint Condition [B.C.08]:	6 Satisfactory
(600) UW Substructure Condition [B.C.15]:	N N/A (NBI)	(072) Appr Roadway Align Appraisal [B.AP.01]	8 Equal Desirable Crit
(061) Channel Condition [B.C.09]:	7 Minor Damage	(071) Waterway Adequacy [B.AP.02]:	8 Equal Desirable
(601) Channel Protection Condition [B.C.09]:	N N/A (NBI)	(113) Scour Condition:	8 - Calc Scour Above Ftg
(633) NSTM Condition:	N/A (NBI) (No NSTMs)	(631) Scour Condition Rating:	8 - Insignificant

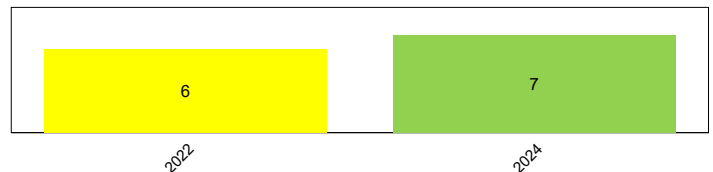
GEOMETRY

(034) Skew [B.G.11]:	0	(032) Approach Roadway [B.G.09]:	30.0
(035) Structure Flared:	0 No flare	(051) Curb To Curb Width [B.G.06]:	38.0
(049) Structure Length [B.G.01]:	600.0	(052) Deck Width Out-To-Out [B.G.05]:	41.1
(619) Curved Bridge [B.G.12]:	N - Not Curved	(441) Deck Area [B.G.16]:	24,650.00
(620) Maximum Height Bridge [B.G.13]:	11	(033) Bridge Medians [B.G.10]:	0 No median
(621) Side Hill Bridge [B.G.14]:	No	(50A) Curb Sidewalk Width/Left [B.G.07]:	0.0
		(50B) Curb Sidewalk Width/Right [B.G.08]:	0.0
		(606) Minimum Span Length [B.G.04]:	40.0

DECK

Deck Material and Type

(501) Deck Overburden Depth (in):	4.0
(107) Deck Type:	2 - Conc Precast Panel
(108A) Wearing Surface:	2 Integral Concrete
(108B) Membrane:	0 - None
(108C) Deck Protection:	0 None



DECK CONDITION

Roadside Hardware

(36A) Bridge Rail:	0 Substandard
(36B) Transition:	1 Meets Standards
(36C) Approach Rail:	1 Meets Standards
(36D) Approach Rail Ends:	N N/A or not required



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SUPERSTRUCTURE

Main Span Material and Type

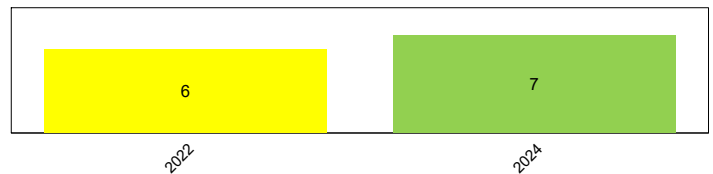
(045) # of Main Spans: 15
(048) Max Span Length [B.G.03]: 40.0
(43A) Main Material: 5 Prestressed Concrete
(43B) Main Design: 05 Multiple Box Beam
(607) Main Number Of Beam Lines [B.SP.03]: 5
(608) Main Span Continuity [B.SP.05]: 1 - Simple / Single Span
(609) Main Protection System [B.SP.07]: Unprotected Span
(610) Main Deck Interaction [B.SP.08]: IM - Integral/Monolithic
(611) Main Deck Reinforcement Protect [B.SP.12]: 0 - No Protection
(612) Main Deck Stay-In-Place Forms [B.SP.13]: 0 - None

Approach Span Material and Type

(046) # of Approach Spans: 0
(308) Appr Max Span Length: 0.0
(44A) Approach Span Material: 0 N/A (or Other)
(44B) Approach Span Design: 0 N/A (or Other)
(613) Appr Number Of Beam Lines [B.SP.03]: 0
(614) Appr Span Continuity [B.SP.05]: N/A (NBI) No Approach Sp
(615) Appr Protection System [B.SP.07]: N/A (NBI) No Approach Sp
(616) Appr Deck Interaction [B.SP.08]: N/A (NBI) No Approach Sp
(617) Appr Deck Reinforcement Protect [B.SP.12]: N/A (NBI) No Approach Sp
(618) Appr Deck Stay-In-Place Forms [B.SP.13]: N/A (NBI) No Approach Sp

SUPERSTRUCTURE CONDITION

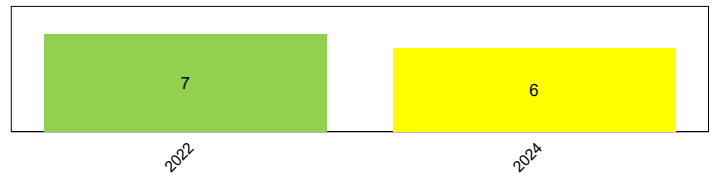
(059) Superstructure Rating [B.C.02]: 7 Good



SUPERSTRUCTURE CONDITION

SUBSTRUCTURE

(060) Substructure Rating: 6 Satisfactory



SUBSTRUCTURE CONDITION

PARKING LOCATION

Parking Latitude (if one provided): 33.894792 Parking Longitude (if one provided): -78.853727
Parking Notes (if provided):
Parked at South shoulder of West approach



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INSPECTION

(090) Date of Inspection: 11/13/2024
(091) Frequency: 24
Water over 4' but UW Inspection Not Needed (No Request): 0
Temperature During Inspection (°F): 61
Weather During Inspection: 1 - Sunny
Team Members: Emmanuel Cooper
Schedule Notes:

Inspection Type	Freq (92)	Last Insp (93)	Next Insp
Routine	24	11/13/2024	11/13/2026
NSTM (A)			
Underwater (B)			
Special (C)			
Other Routine			
Complex Other			
Damage			
Load Rating			

LOAD RATING AND POSTING

(070) Posting %: 5 - Equal/Abv Legal Load
(031) Design Load [B.LR.01]: 9 MS22.5(HS25) or greater

(411) Date Evaluated [B.LR.03]: 03/01/2020
(418A) Deck/Culvert Condition: 8
(418B) Superstructure Condition: 8
(418C) Substructure Condition: 8

(063) Opr Method [B.LR.04]: 6 LFR
(064) Opr Rating [B.LR.06]: 2.63 rf
(065) Inv Method [B.LR.04]: 6 LFR
(066) Inv Rating [B.LR.05]: 1.42 rf
(465) Alternative Inventory Rate Method: 8 LRFR (HL93)
(466) Alternative Inventory Rating Factor: 1.00 rf
(463) Alternative Operating Rate Method: 8 LRFR
(464) Alternative Operating Rating Factor: 1.47 rf

POSTING SIGN VALUES

R-12-6-48 Legal Load Posting and Sign Posted Values

(443) Legal (non-EV) Single 2-3 Axles (tons):
(444) Legal (non-EV) Single 4+ Axles (tons):
(445) Legal (non-EV) Combinations (tons):

R-12-9-36 Emergency Vehicle Posting and Sign Posted Values

(446) EV Single Axle (tons):
(447) EV Tandem Axle (tons):
(448) EV Gross (tons):

40T Combination Posting

(889) 40T Combination Sign Posted: 0 - No, Not Posted

ROADWAY AND CLEARANCES

		Location	
(05B) Kind of Hwy:	3 State Hwy	Route Posted Speed:	-1
(011) Milepost [B.H.07]:	19.887 mi (32.005 km)	(102) Direction of Traffic [B.RT.03]:	1 1-way traffic
(019) Detour Length [B.H.17]:	1 mi (2 km)	SCDOT Route Type:	4
		Evac Route:	1

Clearances

(28A) Lanes On Structure:	2	(622) Railroad Service Type [B.RR.01]:	
(28B) Lanes Under Structure:	0	(574) Railroad Min. Vert. Clearance:	
(053) Min. Vertical Clearance Over Bridge [B.H.13]:	99.99	(575) Railroad Min. Horiz. Offset:	
(054A) Under Vertical Clearance Ref. [B.H.13]:	N Feature not hwy or RR	(038) Navigable Waterway [B.N.01]:	Permit Not Required
(055A) Under Horizontal Clearance Ref. [B.H.15&14]:	N Feature not hwy or RR	(039) Navigation Min Vertical Clearance [B.N.02]:	0.00
(54B) Minimum Vertical Underclearance [B.H.13]:	0.00	(116) Movable Br. Max. Nav. Vert. Clearance [B.N.03]:	0.00
(055B) Minimum Lateral Underclearance R [B.H.15]:	0.00	(040) Navigation Channel Width [B.N.04]:	0.00
(056) Minimum Lateral Underclearance L [B.H.14]:	0.00	(631) Nav. Channel Min. Horiz. Clearance [B.N.05]:	0.00
(010) Wide Load Clearance Vertical:	99.99		



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(047) Wide Load Clearance Horizontal [B.H.16]:

38.00

(111) Substructure Nav. Protection [B.N.06]:

See Inspection > Clearances Page for additional values, OSOW measurements, and notes.

Classification

(026) Funct Class [B.N. 01&02]:

06 Rural Minor Arterial

(109) Pct Trucks [B.H.10]:

9%

(104) NHS:

1 On the NHS

(030) ADT Year [B.H.11]:

2023

(100) Defense Hwy:

0 Not a STRAHNET hwy

(114) Future ADT:

21,297

(029) ADT [B.H.09]:

11,150 Cars/Day

(115) Year Of Future ADT:

2043



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OTHER BRIDGE DATA

(42A) Type Of Service On [B.F.01]:	1 Highway	(506) Signs/Utilities attached:	N - Nothing Attached
(42B) Type Of Service Under [B.F.02]:	5 Waterway	(630) Fatigue Details [B.IR.02]:	
(027) Year Built [B.W.01]:	2000	(507) Damaged Deck:	N - No Damage
(106) Year Reconstructed [B.W.02]:	0	(508) Damaged Superstructure:	N - No Damage
(319) Paint Date:		(509) Damaged Substructure:	N - No Damage
(037) History [B.CL.04]:	5 Not eligible for NRHP	(510) Damaged Culvert:	N - No Damage
(112) NBIS Structure Length:	Y Long Enough		
Status of Consultant Repair Recommendations:	Submitted but Unsigned	(502) Bent or Barrel Direction:	4 - North to South
		(503) Date of Inventory Photos:	12/13/2024
		(504) Asset ID Placard Location:	Bent 1 (NW Corner)
		(505) Stream Flow Direction:	4 - East to West

CURB REVEAL MEASUREMENTS

No Curb:	Yes	Hidden Curb:	No
Curb Reveal Beginning Left (in):		Curb Reveal Beginning Right (in):	
Curb Reveal End Left (in):		Curb Reveal End Right (in):	

CULVERT FILL INVENTORY

Not a Culvert (N/A):	Yes	Max Fill (ft):	
Culvert Inlet (ft):		Culvert Outlet (ft):	
Edge of Roadway on Inlet Side of Road (ft):		Edge of Roadway on Outlet Side of Road (ft):	

ELEMENT CONDITION SUMMARY

Elm/Env	Description	Total Qty	% in 1	Qty. St. 1	% in 2	Qty. St. 2	% in 3	Qty. St. 3	% in 4	Qty. St. 4
15/3	Pre Concrete Top Flange	24,650	100%	24,650	0%	0	0%	0	0%	0
510/3	Wearing Surfaces	24,150	80%	19,311	10%	2,417	10%	2,422	0%	0
3210/3	Del/Spall/Patch/Pot(Wear Surf)	9	0%	0	22%	2	78%	7	0%	0
3220/3	Crack (Wearing Surface)	4,830	0%	0	50%	2,415	50%	2,415	0%	0
104/3	Pre Clsd Box Girder	3,000	99%	2,972	0%	1	1%	27	0%	0
1080/3	Delamination/Spall/Patched Area	28	0%	0	4%	1	96%	27	0%	0
215/3	Re Conc Abutment	84	100%	84	0%	0	0%	0	0%	0
226/3	Pre Conc Pile	56	89%	50	2%	1	9%	5	0%	0
1080/3	Delamination/Spall/Patched Area	6	0%	0	17%	1	83%	5	0%	0
234/3	Re Conc Pier Cap	588	97%	572	0%	2	2%	14	0%	0
1080/3	Delamination/Spall/Patched Area	2	0%	0	0%	0	100%	2	0%	0
1120/3	Efflorescence/Rust Staining	14	0%	0	14%	2	86%	12	0%	0
302/3	Compressn Joint Seal	168	55%	92	28%	47	17%	29	0%	0
2320/3	Seal Adhesion	29	0%	0	35%	10	66%	19	0%	0
2340/3	Seal Cracking	10	0%	0	0%	0	100%	10	0%	0
2350/3	Debris Impaction	37	0%	0	100%	37	0%	0	0%	0
310/4	Elastomeric Bearing	30	100%	30	0%	0	0%	0	0%	0
321/3	Re Conc Approach Slab	1,560	100%	1,560	0%	0	0%	0	0%	0
510/3	Wearing Surfaces	1,520	100%	1,519	0%	0	0%	1	0%	0
3210/3	Del/Spall/Patch/Pot(Wear Surf)	1	0%	0	0%	0	100%	1	0%	0
331/4	Re Conc Bridge Railing	1,200	88%	1,050	13%	150	0%	0	0%	0
1120/4	Efflorescence/Rust Staining	150	0%	0	100%	150	0%	0	0%	0



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ELEMENT NOTES

ELEM/ENV	ELEMENT NAME	QUANTITY	UNITS	QTY ST 1	QTY ST 2	QTY ST 3	QTY ST 4
15/3	Pre Concrete Top Flange	24,650	ft ²	24,650	0	0	0
(15) Spans 40ft each for a total of 600ft; (5) Prestressed Concrete Slab Panels (41ft-1in W x 1ft-6in thick).							
Top of Slab: Unable to inspect due to concrete overlay.							
ELEM/ENV	ELEMENT NAME	QUANTITY	UNITS	QTY ST 1	QTY ST 2	QTY ST 3	QTY ST 4
510/3	Wearing Surfaces	24,150	ft ²	19,311	2,417	2,422	0
Concrete Overlay (600ft L x 40ft-3in W x 4in thick).							
Throughout all spans, longitudinal, diagonal, and map cracking (up to 1/8in W x 20% of surface area). [3220, CS2, Q2415 and 3220, CS3, Q2415] Span 2 left shoulder at BT 2, spall (3ft W x 9in L x 1in deep). [3210, CS3, Q3] Span 7 left shoulder at BT8, spall (2ft-8in W x 1ft L x 1in deep). [3210, CS3, Q3] Span 10 left shoulder at BT10, spall (6in L x 2in W x 1/4in deep). [3210, CS2, Q1] Span 11 left travel lane at BT11, spall (6in L x 5in W x 1-1/2in deep). [3210, CS3, Q1] Span 15 left travel lane at BT15, spall (4in L x 6in W x 1/2in deep). [3210, CS2, Q1]							
ELEM/ENV	ELEMENT NAME	QUANTITY	UNITS	QTY ST 1	QTY ST 2	QTY ST 3	QTY ST 4
3210/3	Del/Spall/Patch/Pot(Wear Surf)	9	ft ²	0	2	7	0
ELEM/ENV	ELEMENT NAME	QUANTITY	UNITS	QTY ST 1	QTY ST 2	QTY ST 3	QTY ST 4
3220/3	Crack (Wearing Surface)	4,830	ft ²	0	2,415	2,415	0



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ELEM/ENV	ELEMENT NAME	QUANTITY	UNITS	QTY ST 1	QTY ST 2	QTY ST 3	QTY ST 4
104/3	Pre Clsd Box Girder	3,000	ft	2,972	1	27	0

(5) Prestressed Concrete Slab Panels; (2) exterior (8ft 6-1/2in W x 1ft-6in thick) and (3) Interior (8ft W x 1ft-6in thick).

Scattered throughout all spans, between beams, efflorescence leaching out.

Throughout all spans, at isolated locations areas of vertical misalignment between beams (up to 1in).

BM 4-2 at at BT4, right edge, spall (6in L x 2-1/2in W x 1/4in deep). [1080, CS2, Q1]

BM 12-5 at 5ft from BT12, left edge, spall (7in L x 1in W x 3in deep). [1080, CS3, Q1]

Throughout all spans over interior bents on West fascia of Beams, spalls/delaminations (up to 6in L x FH x up to 3in deep) with adjacent vertical crack (FH x up to 1/8in W) at the following locations [1080, CS3, Q14]:

BM1-5 at BT 2
BM3-5 at BT 3
BM4-5 at BT 4
BM4-5 at BT 5
BM6-5 at BT 6
BM7-5 at BT 8
BM9-5 at BT 9
BM9-5 at BT 10
BM10-5 at BT 10
BM11-5 at BT 11
BM12-5 at BT 12
BM13-5 at BT 13
BM14-5 at BT 14
BM15-5 at BT 15

Throughout all spans over interior bents on East fascia of Beams, spalls/delaminations (up to 6in L x FH x up to 3in deep) with adjacent vertical crack (FH x up to 1/8in W) at the following locations [1080, CS3, Q12]:

BM2-1 at BT2
BM3-1 at BT3
BM4-1 at BT4
BM4-1 at BT5
BM6-1 at BT6
BM7-1 at BT7
BM8-1 at BT8
BM10-1 at BT10
BM11-1 at BT11
BM12-1 at BT12
BM14-1 at BT14
BM15-1 at BT15

ELEM/ENV	ELEMENT NAME	QUANTITY	UNITS	QTY ST 1	QTY ST 2	QTY ST 3	QTY ST 4
1080/3	Delamination/Spall/Patched Area	28	ft	0	1	27	0

ELEM/ENV	ELEMENT NAME	QUANTITY	UNITS	QTY ST 1	QTY ST 2	QTY ST 3	QTY ST 4
215/3	Re Conc Abutment	84	ft	84	0	0	0

(2) Spill-Through End Bents with Reinforced Concrete Caps (41ft-6in L x 2ft-6in H x 4ft W) on (4) Prestressed Concrete Piles (1ft-6in square) concealed by compacted fill slope protection.

No significant deficiencies noted at time of Inspection.



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ELEM/ENV	ELEMENT NAME	QUANTITY	UNITS	QTY ST 1	QTY ST 2	QTY ST 3	QTY ST 4
226/3	Pre Conc Pile	56	each	50	1	5	0
(14) Interior Bents with (4) Prestressed Concrete Piles (1ft-6in x 1ft-6in) per bent. Pile 2-1, SW corner at cap, spall (7in H x 2in W x 1in deep). [1080, CS3, Q1] Pile 2-2, SW corner at cap, spall (1ft-2in H x 2in W x 1in deep). [1080, CS3, Q1] Pile 2-3, SW corner at cap, spall (7in H x 2in W x 1in deep). [1080, CS3, Q1] Pile 6-2, NE corner at 1ft from ground, spall (14in H x 7in W x up to 2in deep). [1080, CS3, Q1] Pile 12-4, North face at cap, spall (3-1/2in H x 8in W x 1/2in deep). [1080, CS3, Q1] Pile 15-1, SE corner 2ft below cap, spall (5in diameter x 1/2in deep). [1080, CS2, Q1]							
ELEM/ENV	ELEMENT NAME	QUANTITY	UNITS	QTY ST 1	QTY ST 2	QTY ST 3	QTY ST 4
1080/3	Delamination/Spall/Patched Area	6	each	0	1	5	0
ELEM/ENV	ELEMENT NAME	QUANTITY	UNITS	QTY ST 1	QTY ST 2	QTY ST 3	QTY ST 4
234/3	Re Conc Pier Cap	588	ft	572	2	14	0
(14) Interior Bents with Reinforced Concrete Caps (41ft-6in L x 3ft H x 4ft W). Bent 2, North face at West end, void (15in L x 4in H x 1-1/2in deep). [1080, CS3, Q2] Bent 8, underside at Pile 8-3 at closure pour around pile, efflorescence (2ft L). [1120, CS2, Q2] Bent 14, North above Pile 3, efflorescence build-up (7ft L x up to full height). [1120, CS3, Q7] Bent 15, North above Piles 3 and 4, efflorescence build-up (5ft total x up to full height). [1120, CS3, Q5]							
ELEM/ENV	ELEMENT NAME	QUANTITY	UNITS	QTY ST 1	QTY ST 2	QTY ST 3	QTY ST 4
1080/3	Delamination/Spall/Patched Area	2	ft	0	0	2	0
ELEM/ENV	ELEMENT NAME	QUANTITY	UNITS	QTY ST 1	QTY ST 2	QTY ST 3	QTY ST 4
1120/3	Efflorescence/Rust Staining	14	ft	0	2	12	0
ELEM/ENV	ELEMENT NAME	QUANTITY	UNITS	QTY ST 1	QTY ST 2	QTY ST 3	QTY ST 4
302/3	Compressn Joint Seal	168	ft	92	47	29	0
(4) Compression seal joints (41ft-1in L) located both End Bents 1 and 16, and interior BTs 7, and 13. All joints, scattered areas of debris impaction (up to 8ft L). [2350, CS2, Q32] Joint at Bent 1, throughout travel lanes, areas of depressed seal (up to 1-1/2in deep x 5ft L total), throughout both shoulders debris impaction (5ft L total). [2350, CS2, Q5] [2320, CS3, Q5] Joint at Bent 7, in right shoulder, seal cracking and bulging (10ft L), in travel lanes, scattered areas of adhesion failure (greater than 50% adhered). [2320, CS2, Q5] [2340, CS3, Q10] Joint at Bent 13, in both shoulders areas of adhesion failure with seal depressed and bulging (14ft L total). [2320, CS3, Q14] Joint at Bent 16, right travel lane, depressed seal (1/2in deep x 5ft L). [2320, CS2, Q5]							
ELEM/ENV	ELEMENT NAME	QUANTITY	UNITS	QTY ST 1	QTY ST 2	QTY ST 3	QTY ST 4
2320/3	Seal Adhesion	29	ft	0	10	19	0
ELEM/ENV	ELEMENT NAME	QUANTITY	UNITS	QTY ST 1	QTY ST 2	QTY ST 3	QTY ST 4
2340/3	Seal Cracking	10	ft	0	0	10	0



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2350/3	Debris Impaction	37	ft	0	37	0	0

ELEM/ENV	ELEMENT NAME	QUANTITY	UNITS	QTY ST 1	QTY ST 2	QTY ST 3	QTY ST 4
310/4	Elastomeric Bearing	30	each	30	0	0	0
(30) Elastomeric bearing pads (3/4in thick), (1) at each End Bent and (2) at each Interior Bent.							
No significant defects noted at time of inspection.							

ELEM/ENV	ELEMENT NAME	QUANTITY	UNITS	QTY ST 1	QTY ST 2	QTY ST 3	QTY ST 4
321/3	Re Conc Approach Slab	1,560	ft ²	1,560	0	0	0
(2) Reinforced Concrete Approach Slabs (20ft L x 39ft W x 12in thick).							
No significant deficiencies noted at time of inspection.							

ELEM/ENV	ELEMENT NAME	QUANTITY	UNITS	QTY ST 1	QTY ST 2	QTY ST 3	QTY ST 4
510/3	Wearing Surfaces	1,520	ft ²	1,519	0	1	0
Asphalt Wearing Surface (20ft L x 38ft W x 2in thick).							
North Approach Slab, right lane, next to white line at end of approach slab, pothole (6in diameter x 1-1/2in deep) with exposed steel plate (which is embedded in the asphalt). [3210, CS3, Q1]							

ELEM/ENV	ELEMENT NAME	QUANTITY	UNITS	QTY ST 1	QTY ST 2	QTY ST 3	QTY ST 4
3210/3	Del/Spall/Patch/Pot(Wear Surf)	1	ft ²	0	0	1	0

ELEM/ENV	ELEMENT NAME	QUANTITY	UNITS	QTY ST 1	QTY ST 2	QTY ST 3	QTY ST 4
331/4	Re Conc Bridge Railing	1,200	ft	1,050	150	0	0
(2) Reinforced Concrete Bridge Rails (2ft-8in H x 1ft-5in W).							
Throughout all spans, both rails, spaced 2-4ft apart, vertical cracks (FH x HL) some with efflorescence. [1120, CS2, Q150]							

ELEM/ENV	ELEMENT NAME	QUANTITY	UNITS	QTY ST 1	QTY ST 2	QTY ST 3	QTY ST 4
1120/4	Efflorescence/Rust Staining	150	ft	0	150	0	0



Bridge Inspection Report

09698

Team Leader: Colin Redington

Inspection Date: 11/13/2024

(7) Facility Carried: SC 22 EB

(6) Crossing: JONES BIG SWAMP

Inspection Type(s): Routine

(16) Latitude: 33.893483 (17) Longitude: -78.852736

SECTION 4: NON-ELEMENT INSPECTION NOTES

Miscellaneous Notes

N/A

Traffic Signs

(2) Delineators at the North approach; (1) at each corner.

Encroachments

N/A

Waterway and Channel

Waterway Details

- Channel is dry at the time of inspection.

Debris/Obstructions

- No significant obstructions observed in waterway.

Channel Protection

- Both end bents consist of compacted fill at base of end bent caps.

Embankment Deterioration

- EB1 slope protection, West end, starting at EB1, erosion gully (22ft L x up to 3ft W x up to 2ft deep).

Scour

- No significant scour observed since 2008 Routine Inspection.

Approach Roadway Condition

No significant deficiencies noted at time of inspection.

Fender System

N/A

Median and Other Barriers

36A -0: Reinforced concrete barrier rail (2ft-8in H x 1ft-5in W).

36B -1: Metal thrie-beam transitions with wooden posts.

36C -1: "W" beam railing with wooden posts on approaches.

36D -N: Termination ends greater than 75ft away from structure.

NE guardrail transition 20ft from EB1, impact damage (8ft L).

West approach rail, at EB16, spall (8in L x 16in H x up to 2-1/2in deep).

Under Bridge Railings/Barriers

N/A

Curbs and Sidewalks

N/A

Drainage System

(60) PVC Drainage Pipe (6in diameter); (4) per Span along South Curb.

No significant deficiencies noted at time of inspection.

Diaphragms

N/A

Wingwalls

(4) Reinforced Concrete wingwall at all corners.

No significant deficiencies noted at the time of inspection.

(623) Summary of Previously Performed Maintenance Work [B.W.03]

N/A



Bridge Inspection Report

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Team Leader: Colin Redington

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INSPECTION PHOTOS



Photo 1:
North Approach Looking South



Photo 2:
South Approach Looking North



Photo 3:
Backstation (North Approach Looking North)



Photo 4:
Upstation (South Approach looking South)

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Photo 5:
East Profile Looking West



Photo 6:
West Profile Looking East



Photo 7:
Feature Intersected (Looking Upstream) (East)



Photo 8:
Feature Intersected (Looking Downstream) (West)

Team Leader: Colin Redington

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Photo 9:
Asset ID Plaque at Bent 1 (NW Corner)



Photo 10:
Typical Rail Transition (Northwest Shown)



Photo 11:
Typical Bridge Rail (West Shown)



Photo 12:
Concrete Overlay

Team Leader: Colin Redington

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Photo 13:
Typical Joint at End Bent (End Bent 1 Shown)



Photo 14:
Typical Superstructure (Span 1 Shown)



Photo 15:
Typical Slabs over Bent (Bent 10 Shown)



Photo 16:
Typical End Bent (End Bent 1 Shown)

Team Leader: Colin Redington

Inspection Date: 11/13/2024

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Photo 17:
Typical Substructure (Bent 3 Shown)



Photo 18:
Wearing surface, throughout all spans, longitudinal, diagonal, and map cracking (up to 1/8in W x 20% of surface area).



Photo 19:
Wearing surface, Span 2 left shoulder at BT 2, spall (3ft W x 9in L x 1in deep)



Photo 20:
Wearing surface, Span 7 left shoulder at BT8, spall (2ft-8in W x 1ft L x 1in deep).

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Photo 21:

Wearing surface, Span 11 left travel lane at BT11, spall (6in L x 5in W x 1-1/2in deep).



Photo 22:

Throughout all spans over interior bents on East and West fascias of Beams, spalls/delaminations (up to 6in L x FH x up to 3in deep) with adjacent vertical crack (FH x up to 1/8in W).



Photo 23:

Scattered throughout all spans, between beams, efflorescence leaching out.



Photo 24:

BM 12-5 at 5ft from BT12, left edge, spall (7in L x 1in W x 3in deep).

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Photo 25:

Pile 2-2, SW corner at cap, spall (1ft-2in H x 2in W x 1in deep).



Photo 26:

Pile 6-2, NE corner at 1ft from ground, spall (14in H x 7in W x up to 2in deep).



Photo 27:

Pile 12-4, North face at cap, spall (3-1/2in H x 8in W x 1/2in deep).



Photo 28:

Bent 2 Cap, North face at West end, void (15in L x 4in H x 1-1/2in deep).

Team Leader: Colin Redington

Inspection Date: 11/13/2024

(7) Facility Carried: SC 22 EB

(6) Crossing: JONES BIG SWAMP

Inspection Type(s): Routine

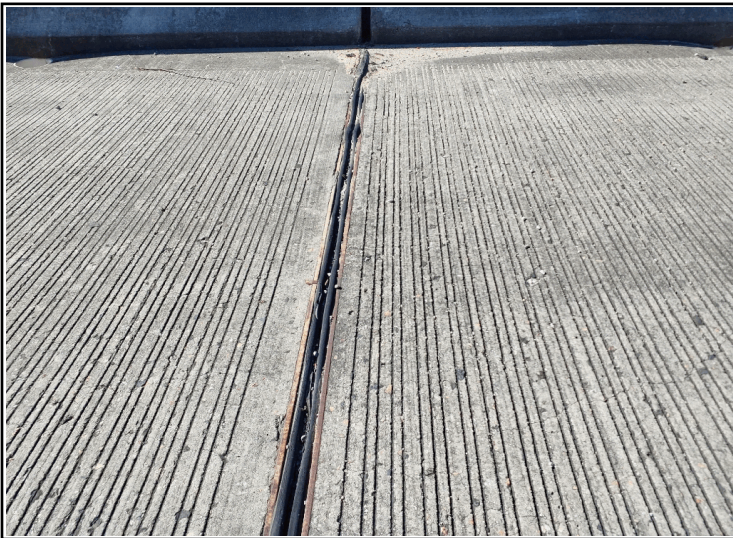
(16) Latitude: 33.893483 (17) Longitude: -78.852736

**Photo 29:**

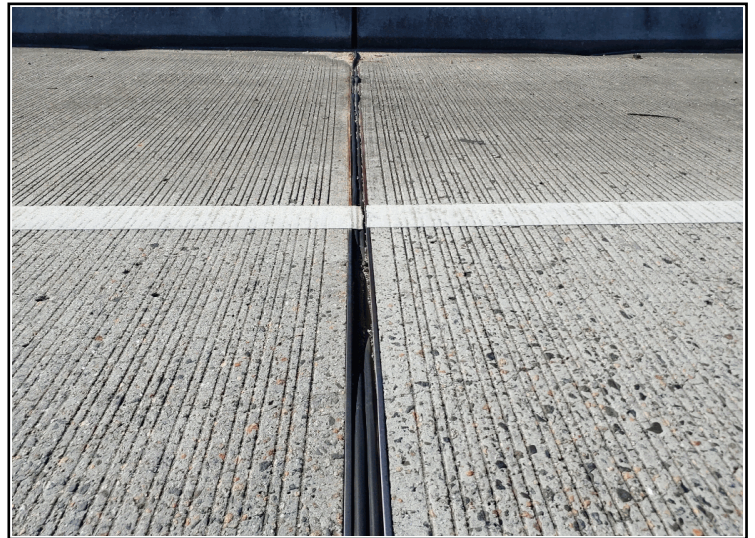
Bent 14 Cap, North face above Pile 3, efflorescence build-up (7ft L x up to full height).

**Photo 30:**

Joint at End Bent 1, throughout travel lanes, areas of depressed seal (up to 1-1/2in deep x 5ft L total), throughout both shoulders debris impaction (5ft L total).

**Photo 31:**

Joint at Bent 7, in right shoulder, seal cracking and bulging (10ft L), in travel lanes, scattered areas of adhesion failure (greater than 50% adhered).

**Photo 32:**

Joint at Bent 13, in both shoulders areas of adhesion failure with seal depressed and bulging (14ft L total).

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Photo 33:

North Approach Slab, right lane, next to white line at end of approach slab, pothole (6in diameter x 1-1/2in deep) with exposed steel plate (which is embedded in the asphalt).



Photo 34:

Throughout all spans, both rails, spaced 2-4ft apart, vertical cracks (FH x HL) some with efflorescence.



Photo 35:

EB1 slope protection, West end, starting at EB1, erosion gully (22ft L x up to 3ft W x up to 2ft deep).



Photo 36:

NE guardrail transition 20ft from EB1, impact damage (8ft L).

Team Leader: Colin Redington

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**Photo 37:**

East approach rail, at EB16, spall (8in L x 16in H x up to 2-1/2in deep).



Bridge Inspection Report

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INSPECTION SKETCHES

Note: Not all bridge inspections have sketches. If no sketches are linked to this report. The image below will read: "Unable to locate graphic."

Unable to locate graphic.



Bridge Inspection Report

09698

Team Leader: Colin Redington

Inspection Date: 11/13/2024

(7) Facility Carried: SC 22 EB

(6) Crossing: JONES BIG SWAMP

Inspection Type(s): Routine

(16) Latitude: 33.893483 (17) Longitude: -78.852736

STREAMBED CROSS SECTIONS

Orientation: Right View
Offset Remark: Downstream Fascia
Elev Basis: Plans
Water Surface:
SND/ELEV Ind.: Soundings

Snding Date: 11/2024
Location of BM: Top of Rail
Offset: 20.42

Streambed Details

Stations	SND/ELEV	Remarks
0 + 0.0	3.6	End Bent 1
0 + 40.0	12.1	Bent 2
0 + 80.0	12.5	Bent 3
1 + 20.0	12.5	Bent 4
1 + 60.0	12.3	Bent 5
2 + 0.0	12.4	Bent 6
2 + 40.0	12.1	Bent 7
2 + 80.0	12.1	Bent 8
3 + 20.0	11.8	Bent 9
3 + 60.0	11.9	Bent 10
4 + 0.0	12.0	Bent 11
4 + 40.0	11.7	Bent 12
4 + 80.0	11.9	Bent 13
5 + 20.0	12.1	Bent 14
5 + 60.0	11.1	Bent 15
6 + 0.0	3.6	End Bent 16

STREAMBED CROSS SECTIONS

Orientation: Left View
Offset Remark: Upstream Fascia
Elev Basis: Plans
Water Surface:
SND/ELEV Ind.: Soundings

Snding Date: 11/2024
Location of BM: Top of Rail
Offset: 20.42

Streambed Details

Stations	SND/ELEV	Remarks
0 + 0.0	3.4	End Bent 1
0 + 40.0	12.3	Bent 2
0 + 80.0	13.0	Bent 3
1 + 20.0	13.2	Bent 4
1 + 60.0	13.2	Bent 5
2 + 0.0	13.2	Bent 6
2 + 40.0	13.0	Bent 7
2 + 80.0	12.7	Bent 8
3 + 20.0	13.6	Bent 9
3 + 60.0	13.3	Bent 10
4 + 0.0	13.1	Bent 11
4 + 40.0	12.9	Bent 12
4 + 80.0	12.5	Bent 13
5 + 20.0	12.4	Bent 14



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Team Leader: Colin Redington
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5 + 60.0	12.4		Bent 15
6 + 0.0	3.9		End Bent 16



Bridge Inspection Report

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SCOUR AND WATERWAY

Request Scour Evaluation:	No	Request Waterway Adequacy:	No
<u>Request Scour Evaluation Reason:</u>		<u>Request Waterway Adequacy Evaluation Reason:</u>	
Inspection Information		Streambed Information	
(511) Bridge Over Tidal Waterway:	No	(519) Water Level:	1 - Within Banks
(512) Tide Status During Inspection:		(520) Debris Buildup:	No
(513) Time During Tidal Inspection:		(521) Bed Deposits Downstream:	No
(517) Max Water Depth at Bent (ft)	0.0	(522) Up/Dwnstrm Ch. Stab.:	Channel Stable
(518) Dry Channel:	Yes	(523) Riprap (if any) Displaced:	N/A (No Riprap)
		(524) Structure on a Channel Change:	No
		(525) Stream Aggradation or Degradation:	None Observed
		(526) Lateral Movement of Stream:	No Movement
		(527) Lakes, dams near the bridge?	No Feature
Evidence of Scour		Underwater Bridge Inspection Only	
(528) Scour Holes at End Bents:	Yes	(534) SSU's Inspected:	
(529) Scour Holes at Bents:	No		
(530) Scour Holes at Culvert Inlet:	No		
(531) Scour Holes at Culvert Outlet:	No	(535) ADCI Dive Supervisor:	
(532) Footing Exposed:	No	(536) ADCI Tender/Diver(s):	
(533) Bank Soil Material Washing Out:	Yes		
Scour Condition, POA and Vulnerability			
(442A) Scour Plan of Action Followed:	No	(537) Water Temperature:	
(442B) POA Category 1:		(538) Visibility:	
(442C) POA Category 2:		(539) Flow Velocity:	
(450) Scour Risk Bin:			
(451) Monitoring / Scour Risk Bin:			

BRIDGE MAINTENANCE OFFICE FIELDS

(099) Border Bridge Structure Number:	(SBI) Bridge with Complex Feature:	0
(098) Border Bridge % Responsible:	(312) Responsible District:	5
(98A) Border Bridge FHWA Region:	(311) Responsible County:	26
(98A) Border Bridge State:	(439) Asset Management:	0
(101) Parallel Structure:	(412) Critical Security bridge (CSB):	No
(103) Temporary Structure:	(SBI) Topside Inspection Category:	SB5
(422) Date of Temporary Structure:	(SBI) Underwater Inspection Category:	
(SBI) Island Mobility Bridge:		

JOINT

Bent Number	Joint Type	Joint Opening at Left Gutter	Joint Opening at Centerline Road	Joint Opening at Right Gutter

Notes:



Bridge Inspection Report

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BEARINGS						
Bent #	Bearing #	Bearing Type	Rotational movt. (°)	Longitudinal movt. (in)	Transverse movt. (in)	Vertical movt. (in)
Notes:						

PILES							
Bent #	Pile #	Pile Material	Splice	Stud-Up	Pile % Rem	Pile Marked	Pile Size / Thickness
Notes:							



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PROCEDURES FOR THIS INSPECTION

Done	Insp Type	Procedure Type	Procedure Name	Details
S	Routine	Routine (Typical)	Inspection: Posting Photos	Photographs of at bridge and advance (if present) posting or weight limit signs are required to be taken during all routine inspections. These requirements are applicable to the posting or weight limit signs at the bridge and in advance of the bridge which could require many signs to be photographed. These signs include R-12-6-48 and R-12-7-48 (legal loads), and R-12-9-36 (EV weight limit) at both ends of the bridge. Per BIGD 5.4.4.2 (TN03).
R	Routine	Routine (Typical)	Post-Inspection: Requests	Confirm that all Inspection Requests are included as both scheduled inspections in the Schedule Table and the Requests Table when needed. Confirm requests include Load Rating (if needed) (NBI GCR 7 to 5 or NBI GCR to 4, Element CS4, etc.). Confirm requests include Hydraulic Appraisal (if needed) (NBI 071 and NBI 113). Confirm requests include UW (Unscheduled) (Water Depth 4' or more at Bent).
S	Routine	Routine (Typical)	Inspection: Timber Sounding	Sound full length of exposed timber piles (including through bolts) per BIGD 5.3.3.1 and 5.3.3.2.1.
R	Routine	Routine (Typical)	Inspection: Defect Photos	Any NBE (element) with CS3 or CS4 Any deterioration causing an NBE (element) to have a CS3 or CS 4 documented with a photograph. Per BIGD 5.4.4.2.



Bridge Inspection Report

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R	Routine	Routine (Typical)	Post-Inspection: Data Verification	Verify the correct: 1) Traffic Status (NBI 041), 2) Inspection Intervals (NBI 090 to 093), 3) Bridge Condition (NBI 58, 59, 60 or 62), 4) Rating Values (NBI 064 and 066), and 5) Sign Posting Values (SBI 443 to 448 and 889).
S	Routine	Routine (Typical)	Inspection: Pile Sketch	Pile Sketch Table required when NBI Item 60 Condition Rating = 5. Per BIGD 5.3.3.1 and 5.4.4.5.

Inspector Procedure Notes
Substructure Rating greater than 5, No Pile Sketch Needed; No Timber Piles; Bridge is not Posted

EQUIPMENT USED IN THIS INSPECTION

<u>Name</u>	<u>Hours</u>	<u>Cost</u>
<u>Inspector Equipment Notes</u>		



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INSPECTOR SIGNATURE AND QC/QA INFORMATION

INSPECTED BY Colin Redington



Date 11/13/2024

QC Reviewed By: Ramesh Mishra QC Completed: 12/31/2024

QA Reviewed By: QA Completed: